MANAGEMENT PORTFOLIO

INFERNO RACING

DESIGN.
DEVELOP.
DOMINATE.

Ansys · ROYAL ENFIELD · Lenovo · SHRI Agrawal Sweets & Namkeen SINCE 1956 · AiREN INTERNATIONAL WORLD-CLASS PACKAGING · THE SHISHUKUNJ INTERNATIONAL SCHOOL INDORE · zevar by SETHIS · tekzee technologies · CHOPADA MOTORS · BALAJI GROUP PROTECTING PEOPLE AND PLACES · raah foundation

TEAM IDENTITY

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Project Title	Inferno Racing IR-11 — STEM Racing India, Season 08
Team	5 members — Design, Manufacturing, Aerodynamics/Computational, R&D, Sponsorship & Marketing
Total Deliverables	8 — two cars, three portfolios, engineering drawings & renders, pit display, verbal presentation
Project Budget	₹30,000 (project spend, excl. travel & logistics)
Project Start	Nov 11, 2025
Regional Competition	3 April 2026 (qualified for Nationals)
National Finals	9 August 2026 — Chennai
Scope	Full CAD / CFD / CAM design cycle, custom wheel & tether-guide manufacture, all portfolios, pit display, verbal presentation

KICK-OFF & PROJECT CHARTER

Inferno Racing officially began on 11 November 2025, when all five of us sat down together at Shishukunj International School for our kick-off meeting. Everyone had been reading the regulations on their own before that; the meeting was where all of that loose research finally turned into a plan we could act on.

We ran the team without a hierarchy. There was no captain making calls over everyone's head — every major decision, from picking the final CFD concept to how we split the budget, was made together and only moved forward once all five of us agreed. What we did do at kick-off was assign responsibilities by strength rather than handing them out evenly, because at Regionals we learned the hard way that splitting work equally usually means people end up doing the things they're weakest at.

PROJECT CHARTER

Straight after kick-off we wrote a full Project Charter and all five of us signed it. It fixed our deliverables, budget, scope and milestones, and stayed the reference document we came back to whenever a decision needed settling.

SCOPE STATEMENT

We defined scope tightly at the start to keep scope creep out. In scope: the full CAD/CFD/CAM cycle, custom wheel and tether-line guide manufacturing, all three competition portfolios, engineering drawings and 3D renders, the pit display, sponsor acquisition, and verbal presentation prep. Explicitly out of scope: any post-submission modification to the car, which parc fermé rules prohibit anyway.

"Lights Out" — Reaction Trainer

In this class of racing the result is the reaction plus the car. So we built our own tool to train the reaction — a web-based F1 start-lights simulator the whole squad uses to measure and shave down reaction time.

How it works

Five lights come on one at a time at irregular intervals, so you can't count a rhythm. Once all five are lit they hold for a random spell, then go dark at once — you react the instant they go out. Press during the hold and it's logged as a jump start, exactly like a real false start off the line.

The engineering behind it

Memoryless hold. The hold time is drawn from an exponential distribution, so how long you've already waited tells you nothing about when the lights will drop. Guessing has no payoff — the tail runs out to 8 seconds so long holds stay honest.

Frame-accurate timing. The trainer measures the display's refresh rate and times reaction from lights-out, doing zero rendering at the moment of lights-out so nothing on screen can tip you off early or bias the measurement.

Median over best. A single quick rep is luck. The dashboard reads the median and the slowest 10% to show real, repeatable reaction; a genuine personal best shows in green.

Squad sync. Runs sync to a shared team sheet through a Google Apps Script endpoint, with an offline queue so nothing is lost, feeding a live squad log for all five athletes.

Gamepad support. Works with a controller, so we train on the same kind of button we'd actually use, not just a keyboard.

5 ATHLETES TRACKED
UV · ANAY · DAKSH · ANSH · VISHESH

8S EXPONENTIAL HOLD TAIL — KEEPS GUESSING POINTLESS

±½ FRAME-LEVEL TIMING PRECISION (REFRESH-AWARE)

Why it matters for the portfolio: it's evidence of the team applying real engineering thinking — probability, timing systems, data sync — to a performance problem we own, and it gives us objective numbers to improve against instead of guesswork.

ROLES

ROLES & RESPONSIBILITIES

Rather than a manager sitting above everyone, we split the team into five functional leads and made each person the owner of their area. Everyone still backs each other up when a deadline gets tight, but ownership is always clear.

Member	Role	Owns
Yuvraj Nikum (UV)	Marketing Manager & Manufacturing	Branding, social media, CAM toolpaths, CNC liaison, assembly, QA & regulation compliance
Ansh Agrawal	Sponsorship Manager	Sponsor outreach, proposals, partnerships, enterprise portfolio
Daksh Firoda	R&D Engineer	Design research, component testing, performance analysis, benchmarking
Anay Mehta	Design Engineer	CAD modelling, design iterations, engineering drawings, 3D renders
Vishesh Dagra	Computational Engineer	CFD setup, simulation runs, results analysis, wing optimisation

RACI Matrix

Because we don't run a single project manager, accountability is spread across the leads. Coordination work — planning, the schedule, the portfolios and the verbal presentation — is jointly owned by all five of us, and budget is jointly owned by Marketing and Sponsorship. Everything else sits with whoever leads that function.

A — Accountable (final ownership) · R — Responsible (does the work) · C — Consulted · I — Informed

Deliverable	UV·Mktg	Ansh·Spon	Daksh·R&D	Anay·Design	Vishesh·Comp
Project Planning	A	A	A	A	A
Timeline & Milestones	A	A	A	A	A
Research & Benchmarking	I	I	A/R	C	C
Regulation Compliance	A/R	I	C	C	C
CAD Development	I	I	C	A/R	C
CFD Analysis	I	I	C	C	A/R
Manufacturing	A/R	I	C	C	C
Testing & Validation	I	I	A/R	C	R
Design Iterations	I	I	C	A/R	R
Sponsorship Acquisition	C	A/R	I	I	I
Budget Management	A	A	I	I	I
Marketing & Branding	A/R	C	I	I	I
Social Media	A/R	C	I	I	I
Portfolio Development	A	A	A	A	A
Verbal Presentation	A	A	A	A	A

ROLES & SCHEDULE

WORK BREAKDOWN STRUCTURE

We broke the project into five work streams, one per member, then decomposed each into its own tasks. Matching streams to people meant every task had a clear owner from day one.

Work Stream	Key Tasks	Owner
Design & CAD	Concept development, CAD modelling, design iterations, engineering drawings, 3D renders	Anay Mehta
Computational	CFD setup, simulation runs, results analysis, wing optimisation, concept selection, re-validation	Vishesh Dagra
Manufacturing	CAM toolpaths, machinist liaison, CNC scheduling, assembly, finishing, QA & compliance	Yuvraj Nikum
R&D & Testing	Design research, component testing, performance analysis, post-regional evaluation	Daksh Firoda
Enterprise & Marketing	Sponsor outreach, proposal decks, brand assets, social media, enterprise portfolio, pit display	Ansh Agrawal

PROJECT SCHEDULE

Our season ran from registration on 15 October 2025 to the National Finals on 9 August 2026 — just under ten months. It didn't run in a straight line, though. It ran in three phases split by a gap we didn't control.

Milestone	Date	Status
Team registration	Oct 15, 2025	Complete
Kick-off & Project Charter signed	Nov 11, 2025	Complete
Design freeze — final CAD approved	Jan 5, 2026	Complete
Regional competition	Apr 3, 2026	Complete
National qualification confirmed	May 30, 2026	Complete
Modified cars received from machinist	Jul 19, 2026	Complete
Internal portfolio deadline	Jul 31, 2026	Complete
Official portfolio submission	Aug 4, 2026	Pending
National Finals	Aug 9, 2026	Pending

Phase 1 — Regional Campaign

11 Nov 2025 – 3 Apr 2026 · 20 weeks

Kick-off, sponsor outreach, research, CAD of concepts A and B, CFD selection, design freeze, CAM, CNC machining of both cars, drawings and renders, assembly and regulation checks, all three portfolios, pit display, and presentation rehearsal — closing at the Regional competition.

Phase 2 — Results Hold

3 Apr – 30 May 2026 · 8 weeks

No project work was possible here. We couldn't modify the car without knowing whether we'd qualified, and couldn't justify spending the rest of the budget on machining until we did. We've documented this phase rather than hidden it — it's the single biggest constraint on our whole schedule. Eight of our ten available months were gone before the National campaign could even start.

Phase 3 — National Campaign

30 May – 9 Aug 2026 · 10 weeks

Qualification confirmed, performance review and speed R&D, CAD revision of the modified geometry, CFD re-validation, CAM to the machinist, CNC of the modified cars, then assembly, finishing and regulation checks — with portfolio rewriting and presentation prep running in parallel the whole time.

Critical Path & Float

Our critical path for Nationals runs: qualification result → performance review → CAD revision → CFD re-validation → CAM toolpaths → CNC machining → assembly & regulation checks → submission. Adding those durations gives roughly nine weeks of critical work against the nine and a half weeks available before the 4 August deadline.

Total float across the National campaign ≈ 4 days

This is the most important number in our schedule, and we want to be direct about it — it isn't comfortable. It's the direct result of an eight-week external hold we could neither shorten nor work through, which squeezed a full design-and-manufacture cycle into ten weeks. Knowing the figure is exactly why we chose to write the portfolios in parallel rather than in sequence, and why we set our own internal deadline earlier than the official one.

SCHEDULE

SCHEDULE VARIANCE & BUFFER

Two Phase-1 tasks ran late, both logged in our status reports. Regional CNC machining slipped one week (planned 1 Feb, delivered 8 Feb) because our Indore machinist had jobs queued ahead of ours; with no realistic backup at that budget, we absorbed it into assembly float and it didn't move the competition date. Portfolio drafting started two weeks late (planned 2 Mar, began 16 Mar) when manufacturing overran into the writing window — the failure we most wanted to avoid repeating. So for Nationals we started portfolio revision on 1 June, the day after qualification, and ran it alongside the entire build. Our portfolios are officially due 4 August; we set our own deadline at 31 July, giving four deliberate buffer days to absorb one late problem — a failed check, a corrupted file, an upload issue — without missing the deadline that actually counts.

BUDGET & FINANCIAL MANAGEMENT

Our funding came from sponsorship, calculated across four cash tiers plus in-kind partnerships. Rather than chase one big backer, we built a tiered structure so no single sponsor held up the whole budget.

Tier	Per partner	Partners	Cash
Principal	₹15,000	Royal Enfield · Lenovo	₹30,000
Monetary	₹10,000	Shishukunj · Airen International · Chopda Motors	₹30,000
Aid	₹5,000	Tekzee · Balaji Group · Zewar	₹15,000
Sustainability	in-kind	Raah Foundation	-
Food & Beverages	in-kind	Shri Agrawal Sweets & Namkeen	-
Analysis	in-kind	ANSYS (software licences)	-
Total		11 partners	₹75,000

PROJECT SCHEDULE — GANTT CHART

Task / Milestone	JUNE	JUNE	JULY	JULY	AUG	AUG	Owner	Status
Kick-off & Project Charter	●						All	Done
Sponsor outreach & proposals	●	●					Ansh	Done
Research & concept development	●	●					Daksh	Done
CAD modelling — concepts A & B	●	●					Anay	Done
CFD analysis & concept selection		●					Vishesh	Done
Design freeze — final CAD approved		●					Anay	Done
CAM toolpaths generated			●				Anay	Done
CNC machining — both cars			●	●			Daksh	Done
Engineering drawings & 3D renders				●			Anay	Done
Car assembly & regulation check				●	●		Yuvraj	Pending
All three portfolios — draft & finalise					●		All	In progress
Pit display — design & build					●		Ansh	Done
Verbal presentation — rehearsal					●	●	All	Pending
Competition Day						Aug …	All	—

TRAVEL & LOGISTICS (CHENNAI)

Item	Cost	Notes
Travel	₹85,000	Team travel to Chennai
Accommodation	₹20,000	The Banyan Palace Banquet Hall · 8–10 Aug
Travel total	**₹1,05,000**	Funded separately from project budget

RISK

RESOURCE MANAGEMENT

Physical

- STEM Racing India official model block
- CNC machine (external machinist, Indore)
- Finishing tools, sandpaper, paint
- Decal printing materials
- Pit-display board and print materials

Digital

- Autodesk Fusion 360 — CAD & CAM
- SimScale / ANSYS — CFD analysis
- Canva — portfolio & brand design
- Instagram — digital marketing
- WhatsApp — team communication

Band	Score	Response
Low	0.00 – 0.20	One plan, checked at reviews
Medium	0.21 – 0.40	Target whichever is higher — probability or impact
High	0.41 – 1.00	Two plans: one to avoid, one to reduce the damage

RISK REGISTER — OPPORTUNITIES

#	Opportunity	Score	Plan
O1	Free student software licences	0.48	Exploit — Fusion 360, SimScale & ANSYS at ₹0; that money went into machining
O2	CNC price was negotiable	0.36	Exploit — build down to ₹23,500 against ₹25,000; ₹1,500 back into contingency
O3	Strong local business network in Indore	0.36	Enhance — pitched in person; eleven partners signed, several Indore companies
O4	Sustainability partner in Raah Foundation	0.24	Share — joint STEM outreach & donation drives, strengthening the Enterprise entry

RISK REGISTER — THREATS

#	Risk	P	I	Score	Band	Plan
T1	Result date out of our hands	0.8	0.8	0.64	High	Accept + mitigate — keep the team together, research, and design the day results dropped
T2	Barely any spare time before Nationals	0.6	0.8	0.48	High	Mitigate — write portfolios in parallel with the build, not after
T3	Cars damaged en route to Chennai	0.4	1	0.4	Med	Avoid — hand-carry in foam-lined hard case, spares packed separately
T4	Only one machinist, no backup	0.6	0.6	0.36	Med	Mitigate — send toolpaths early, book the slot ahead, follow up weekly
T5	Parts breaking during testing	0.6	0.6	0.36	Med	Mitigate — machined spares alongside the main build; rules allow replacements
T6	Sponsor money short or late	0.6	0.6	0.36	Med	Mitigate — four tiers, eleven partners; no single sponsor held up the budget
T7	Portfolio writing starting late	0.6	0.6	0.36	Med	Mitigate — started the day after qualification, not after the cars were done
T8	Failing scrutineering on a performance rule	0.4	0.8	0.32	Med	Avoid — measured both cars against the scorecard before travelling
T9	Member unavailable during exams	0.8	0.4	0.32	Med	Mitigate — exam weeks blocked out; overlapping skills so no job depended on one person
T10	CFD results we couldn't trust	0.6	0.4	0.24	Med	Mitigate — fixed mesh settings, re-ran before design freeze
T11	Missing the submission deadline	0.2	1	0.2	Low	Avoid — internal deadline 31 Jul vs official 4 Aug (four spare days)
T12	Losing CAD or portfolio files	0.2	0.8	0.16	Low	Mitigate — cloud storage plus local copies with two members

COMMUNICATION

HOW WE COMMUNICATED

After Regionals we tightened things up — daily WhatsApp check-ins, a weekly review every Sat/Sun, and milestone meetings for bigger calls. Big decisions (CAD, CFD, manufacturing, portfolio) got an informal look 2–3 days early, then a formal sign-off. Every meeting ended with clear action points, an owner, and a deadline, all logged in the tracker.
Sponsors got ROI updates, the machinist was looped in on tolerances before we built anything, and mentors reviewed our engineering calls at key points — feedback always made it into the next week's plan.

Stakeholder	How	How Often	Owner
Team	WhatsApp + weekly meetings	Daily	All
Faculty mentor	Google Meet reviews	As needed	All
Sponsors	In-person, email, phone	Outreach + updates	Ansh
CNC manufacturer	Calls, CAD reviews	As needed	UV & Daksh
Technical advisors	Online meetings	Key milestones	Vishesh
School admin	Letters, meetings	Monthly	Ansh
Public	Instagram	Regular	UV

HOW WE TRACKED PROGRESS

We checked everything against schedule, budget, technical goals, and scope continuously, not just at checkpoints. Whenever something drifted, we ran the same loop:

Measure→Analyse→Decide→Implement→Verify→Standardise

Any proposed change got checked against technical performance, manufacturing feasibility, schedule, and budget — only clear net wins got approved.

STATUS REPORTS

Report	When	Issue	Fix
SR-01 Initiation	Nov 2025	None	Sponsor deck done, outreach started
SR-02 Design	Dec 2025	Mesh too coarse	Adjusted, re-ran CFD before freeze
SR-03 Manufacturing	Jan 2026	Cars delayed	Weekly follow-ups, portfolio work moved forward
SR-04 Portfolio	Mar 2026	PM portfolio started late	Dedicated March time, full team on prep

KPIS

KPI	Target	Actual	Status
Sponsors	5+	11 across 6 tiers	Exceeded
Sponsorship raised	—	₹75,000 cash + in-kind	Exceeded
Budget adherence	Within ₹30,000	₹23,500 spent	On track
CNC machining cost	₹25,000	₹23,500	Under
Instagram followers	100+	146	Exceeded
Instagram reach (30d)	10,000+	27,700	Exceeded
Portfolios completed	3 of 3	Final revision	In progress
Regulation compliance	Full	Scorecard-checked	Verifying
Deliverables on time	100%	On track for 9 Aug	On track

SCOPE & OVERALL HEALTH

No formal scope changes all season — stayed inside the original Charter. The one big call, a custom wheel and tether-line guide over stock parts, was locked in early and held. Project health: GREEN